Soumyadeep Chandra

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Research Focus

Ph.D. researcher in Electrical and Computer Engineering at Purdue University focused on **semiconductor thermal modeling, 3D heterogeneous integration (3DHI), advanced packaging, and AI for hardware**. Develops high-fidelity **COMSOL finite-element simulation** workflows and physics-informed surrogate models for fast steady-state and transient chip-temperature prediction. Background in **VLSI design, circuit power and thermal analysis, and physics-informed machine learning**. **Expected graduation: Spring 2027.**

Education

Purdue University **Jan. 2021 – Present**
Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Kaushik Roy *GPA: 3.88/4.00*
Coursework: AI for Hardware, MOS VLSI Design, Advanced VLSI, Algorithms, Deep Learning, Reinforcement Learning

Jadavpur University **Aug. 2016 – May 2020**
B.E. (Hons.) in Electronics and Telecommunication Engineering *GPA: 9.48/10.00*

Technical Skills

Modeling: 3D Heterogeneous Integration (3DHI), Thermal Management, Finite Element Method (FEM), Steady-State and Transient Thermal Analysis, Power and Hotspot Modeling

Simulation & EDA Tools: COMSOL Multiphysics, Cadence Celsius, Cadence Virtuoso, HSPICE, Verilog

Programming: Python, MATLAB, Git, Bash, C, C++, Linux, Numerical Modeling

Machine Learning: PyTorch, TensorFlow, Neural Operators, Physics-Informed ML, Deep Learning

Research Experience

Nanoelectronics Research Lab, Purdue University **West Lafayette, IN**
Graduate Research Assistant *Jan. 2023 – Present*

Thermal Modeling & Analysis

[1] **Therm3D – Physics-Aware Neural Operator for Transient Thermal Modeling in 3DHI** (*Under Review*)

- Developed a physics-informed neural operator framework for near-real-time **steady-state and transient thermal analysis** of 3D heterogeneous integration (3DHI) systems.
- Built an automated COMSOL–Python FEM and synthetic data generation pipeline and developed a point-cloud latent **neural operator** with coordinate-conditioned decoding and multi-stage autoregressive rollout for long-horizon 3D thermal prediction; incorporated **energy conservation, boundary-condition enforcement, and source/hotspot-aware modeling** for improved physical consistency.
- Achieved **1.09°C RMSE** on unseen power maps over transient thermal trajectories while delivering approximately $\mathbf{10,000 - 20,000\times}$ **faster** inference than FEM, enabling rapid evaluation of 3DHI power and package configurations.

[2] **2D-ThermAI – Physics-Informed Framework for Thermal Analysis of Circuits** (*IEEE TCAD, 2025*)

- Developed a physics-informed deep-learning framework for fast **steady-state and transient thermal prediction** of integrated circuits, combining circuit-level power analysis with high-fidelity COMSOL FEM simulations.
- Built a dataset of **200+ circuit configurations**, from logic gates to complex workloads, and developed a **hybrid U-Net** with positional embeddings, feature-level fusion, and physics-informed regularization to preserve thermal diffusion behavior and spatial consistency.
- Achieved **0.71°C RMSE** with approximately **200× faster inference than COMSOL FEM**; demonstrated generalization across an extended **25 – 95°C** operating range with full-scale RMSE below **2.2%**.

Video & Scene Understanding

[3] **REMAP – Regularized Matching and Partial Alignment of Video Embeddings:** (*Under Review*) (*arXiv*)
Developed a weakly supervised procedure-learning framework using **regularized partial Gromov–Wasserstein optimal transport** to discover temporal correspondences across unaligned instructional videos, enabling robust alignment despite missing, reordered, or unmatched steps.

[4] **OPEL – Optimal Transport for Procedure Learning:** (*NeurIPS 2024*)
Co-developed an **optimal-transport-guided procedure-learning framework** to align temporally unstructured video demonstrations and learn step-level representations across first- and third-person viewpoints.

[5] **ViPeR – Video Representation Learning:** (*IEEE Access, 2025*)
Developed a self-supervised video representation-learning approach for capturing **temporal structure and action-level information** from unlabeled videos, supporting downstream video-understanding and recognition tasks.

Semantic & Syntactic Understanding

[6] **Image Grammar** – **Structured Visual Understanding:** (*IEEE Access, 2024*)
Developed a structured visual-reasoning framework for modeling **compositional, semantic, and syntactic relationships** among objects in complex scenes, enabling representations that capture both visual concepts and their contextual relationships.

[7] **Loss Knows Best** – **Detecting Annotation Errors** (*Under Review*) (*arXiv*)
Developed a learning-based approach for identifying **mislabeled and noisy training samples** by exploiting model-loss behavior, improving annotation-error detection and the reliability of supervised training data.

Publications

[1] **S. Chandra**, S. S. Chowdhury, and K. Roy, "**Therm3D: Physics-Aware Neural Operator for Transient Thermal Modeling in 3D Heterogeneous Integration Systems**," *Under Review at DATE*, 2026.

[2] **S. Chandra**, S. S. Chowdhury, and K. Roy, "**2D-ThermAI: Physics-Informed Framework for Thermal Analysis of Circuits using Generative AI**," *IEEE TCAD*, 2025.

[3] **S. Chandra** and K. Roy, "**REMAP: Regularized Matching and Partial Alignment of Video Embeddings**," arXiv:2509.24382, *Under Review at NeurIPS,* 2026.

[4] S. S. Chowdhury, **S. Chandra**, and K. Roy, "**OPEL: Optimal Transport Guided ProceduрE Learning**," *Advances in NeurIPS*, 2024.

[5] **S. Chandra**, S. S. Chowdhury, C. Yong, C. P. Sundaram, and K. Roy, "**ViPeR: Vision-based Surgical Phase Recognition**," *IEEE Access*, 2025.

[6] S. S. Chowdhury, **S. Chandra**, and K. Roy, "Towards **Visual Syntactical Understanding**," *IEEE Access*, 2024.

[7] P. Alwis*, **S. Chandra***, D. Ravikumar, and K. Roy, "**Loss Knows Best: Detecting Annotation Errors in Videos via Loss Trajectories**," arXiv:2602.15154, *Under Review at WACV*, 2026.

[8] **S. Chandra**, R. Chatterjee, Z. T. Olmsted, A. Mukherjee, and J. L. Paluh, "Axonal Transport During Injury on a Theoretical Axon," *Frontiers in Cellular Neuroscience*, 2023.

[9] P. Mazumder, **S. Chandra**, S. Rana, M. Mukhopadhyay, and M.K.Naskar, "Parallel hardware implementation of Walsh Hadamard transform," *Journal of Scientific & Industrial Research*, 2022.

Teaching Experience

Purdue University — **West Lafayette, IN**
Graduate Teaching Assistant — *2021 – Present*

- **AI for Hardware:** Graduate-level instruction covering machine learning/hardware co-design, AI accelerators, model optimization, and hardware-aware learning.
- **MOS VLSI Design:** Supported graduate-level circuit and layout projects using Cadence, including Wallace-tree multipliers and neuromorphic computing designs.
- **Electrical Engineering Fundamentals:** Served as **Lead Graduate TA** for a large undergraduate course (800+), coordinating a team of graduate and undergraduate teaching assistants.

Prior Work Experience

PricewaterhouseCoopers — **Kolkata, India**
Assistant Consultant — *Aug. 2020 – Dec. 2020*

- Developed frontend and backend enterprise systems using Microsoft Azure for a large retail client.

Jadavpur University — **Kolkata, India**
Research Assistant — *Jan. 2017 – Dec. 2020*

- Developed mathematical models of molecular motor transport in neuronal axons. [8]
- Developed tensor-product-based algorithms for hardware implementation of discrete transforms using FPGA architectures. [9]

Reviewer Service

NeurIPS, AAAI (Program Committee), ICLR, ICML, WACV, MICCAI, TMI, TCDS and ISCAS.

Honors & Awards

- **Finalist, Qualcomm Innovation Fellowship (QIF)**, 2026.
- **Golden Reviewer, ICML**, 2026.
- **Estus H. and Vashti L. Magoon Award for Excellence in Teaching**, Purdue University, 2022.
- **PwC Best Intern Project (Hardware)**, 2019.